Recommendation for Tag Antenna Connection

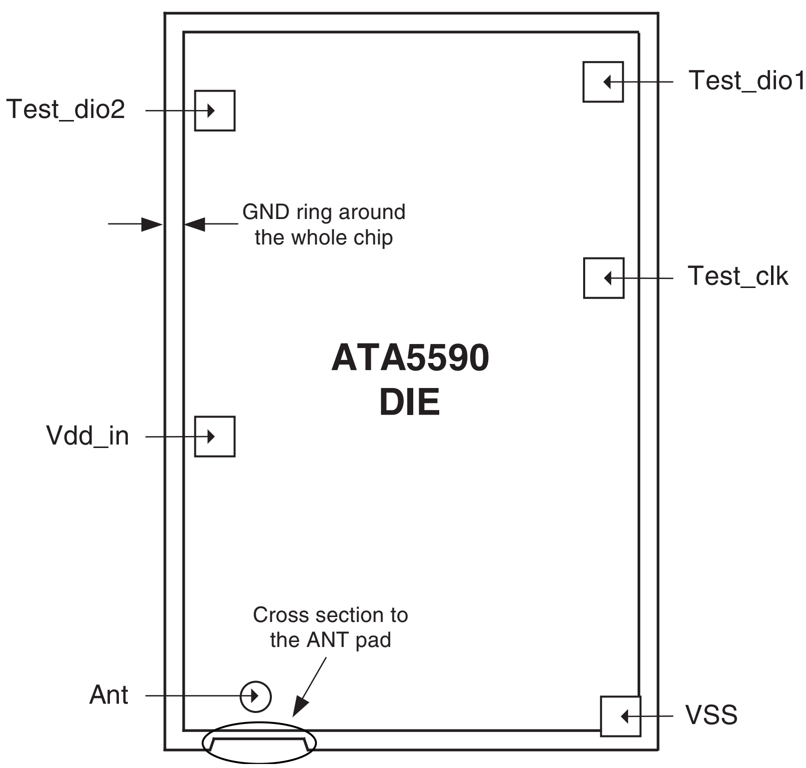
ATA5590 Tag Antenna Connection

Application Note

1. Overview

The connection to the antenna pad should not cross the GND ring surrounding the whole die too much, in order to avoid coupling capacitances. To reduce the parasitic influence of the antenna connection, the GND ring narrows in the antenna area (antenna pad), as can be seen in Figure 1-1 on page 1, and in Figure 1-2 on page 3 with the exact dimensions.

Figure 1-1. Pad Configuration on the Die (View from Top)



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1.1 Electrical Pad Specifications

ESD:	All pads are protected against ESD damage ANT pad: > 1.5 kV (HBM) Other pads: > 1.5 kV (HBM)
Pad connections:	Test_dio2 and Test_dio1 can be connected together; these two pads can also be considered for backup connection in order to improve the mechanical behavior Note: These test pads should not be connected to the ANT pad Test_clk and Vdd_in should not be connected. Note: These two pads are not appropriate for backup connections, and should also not be connected to the ANT pad

1.2 Mechanical Die Specifications

Die dimensions - including nomensland line:	1260 µm × 1890 µm
Die dimensions - excluding nomensland:	1250 µm × 1880 µm
Dimensions of the pads on the die:	Diameter of the ANT pad is 60 µm VSS pad is 86 µm × 114 µm Size of the other pads is 80 µm × 80 µm
Distance between the pads:	Minimum 250 µm (metal edge to metal edge)
Die thickness:	150 µm with bumps 300 µm without bumps
Pad identification - function:	See Table 1-1 on page 3
Die and pad dimensions:	See Figure 1-2 on page 3
Bump location:	See Figure 1-2 on page 3
Passivation material:	Silicon oxide, thickness: 300 µm Silicon nitride, thickness: 750-nm, UV-resistant layer
Passivation thickness:	1050 nm

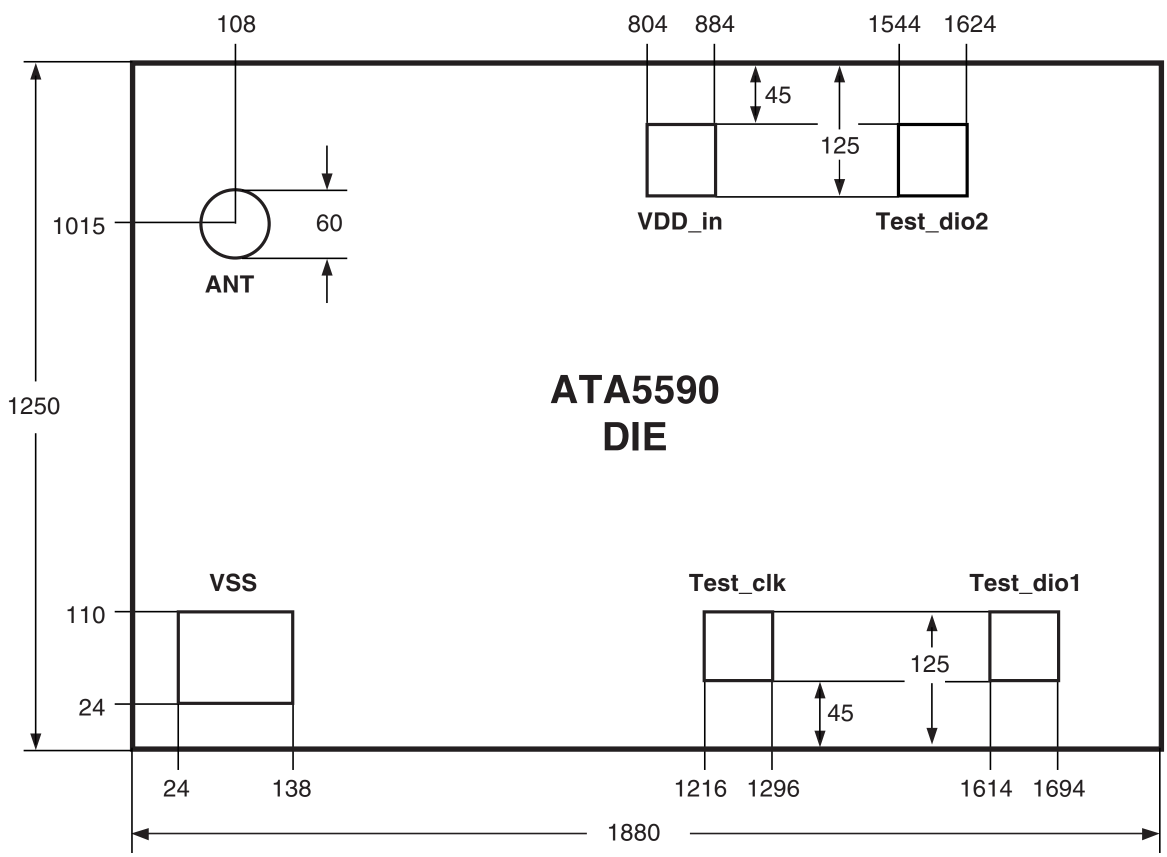
[Table 1-1] lists the pads on the die with their functionality.

Table 1-1. Pads on the Default Die

Pad	Function	Physical Construction	Comment
ANT	Antenna	RF in	Diameter of opening 60 µm
VSS	Antenna_gnd	RF ground	Ground pad
VDD_IN	Test pad	Input/output	VDD after the rectifier
Test_Clk	Test clock in	Input	Notch input
Test_dio1	Test-pad	In_out open drain	Control and observation possibilities
Test_dio2	Test-pad	In_out open drain	Control and observation possibilities

[Figure 1-2] shows the pad dimensions and location on the die.

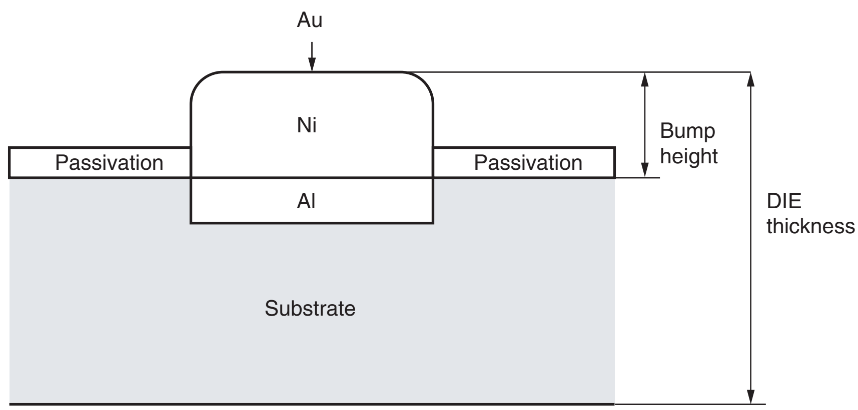
Figure 1-2. Die and Pad Dimensions (in µm)***



1.3 Bump Specifications

Bump material:	Nickel Gold (NiAu) 25-µm Ni thickness 0.1-µm Au thickness (coating) See also Figure 1-3
Bump height:	25 µm (all pads are bumped)
Bump size – Test pads:	80 µm × 80 mm
Bump size – HF pads:	ANT pad: 60 µm VSS pad: 86 µm × 114 µm
Under bump metallization:	Aluminum (Al)

Figure 1-3. Bump – Lateral Cut



1.4 Mechanical Wafer Specifications

Wafer diameter:	6" (150 mm)
Electrical connection of substrate:	VSS
Charge size:	25

1.5 Wafer Delivery Status

Wafer (die) thickness (ground):	150 µm
	300 µm
Failed-die identification:	Each die is electrically tested, failed die are inked

Figure 1-4. Reticle Plan

2. Functional Antenna Connection

In contrast to the VSS pad, the ANT pad significantly influences the read distance of the tag.

Parameters which influence the read distance between reader and tags:

- Tolerance of the connections (ANT pad and VSS pad)
- Coupling capacitances of the cross section of the ANT pad

2.1 Parasitic Capacitance Versus Different Antenna Connections

The antenna connection can be roughly approximated as a plate-type capacitor. Therefore, to calculate the parasitic capacity of the connection, the following formula can be used:

$$C = \varepsilon_r \times \varepsilon_0 \times \frac{A}{d}$$

2.1.1 Sample Parasitic Capacity Calculation

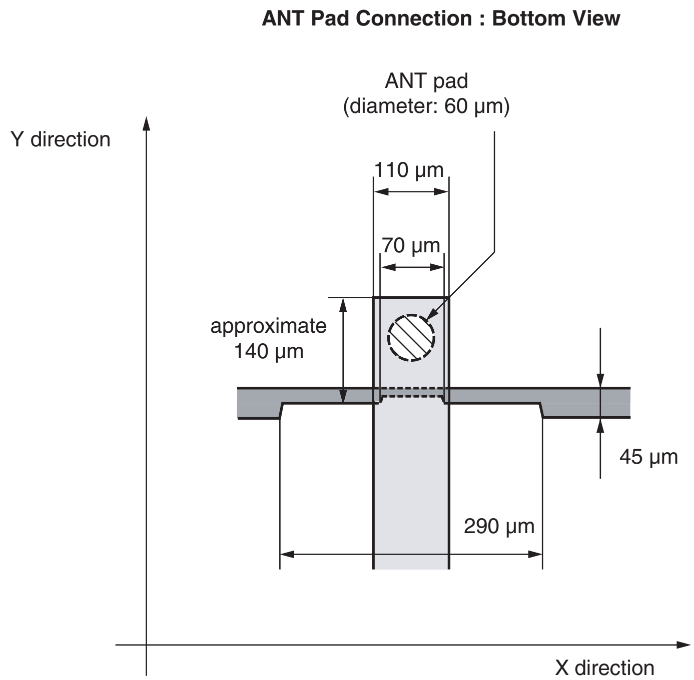
Example: Calculation of the parasitic capacity of the antenna pad connection based on the connection in Figure 2-3 on page 9

Conductive adhesive ε_r:	normally 3 to 4
Dielectric coefficient ε_0:	8.85×10^{-12} F.m^{-1}
Distance d:	approximately 25 µm (bump height)
ANT feed area:	approximately 140 × 110 µm = approximately 15400 µm^2
ANT pad area:	60 µm (diameter) = 2827 µm^2

A = 15400 – 2827 = approximately 12573 µm^2

$$C = 4 \times 8.85 \times 10^{-12} \times \frac{12573\ \mu m^2}{25\ \mu m} = \text{approximately } 17\ \text{fF}$$

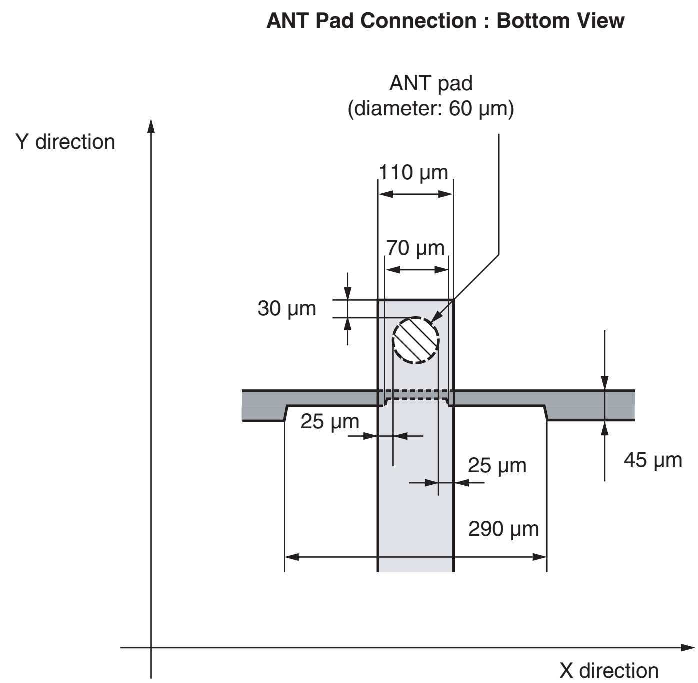
Figure 2-1. ANT Pad Connection Area



2.2 Antenna Connection Recommendation – Assembly Tolerance ±50 µm and an Antenna Feeder Width of 110 µm

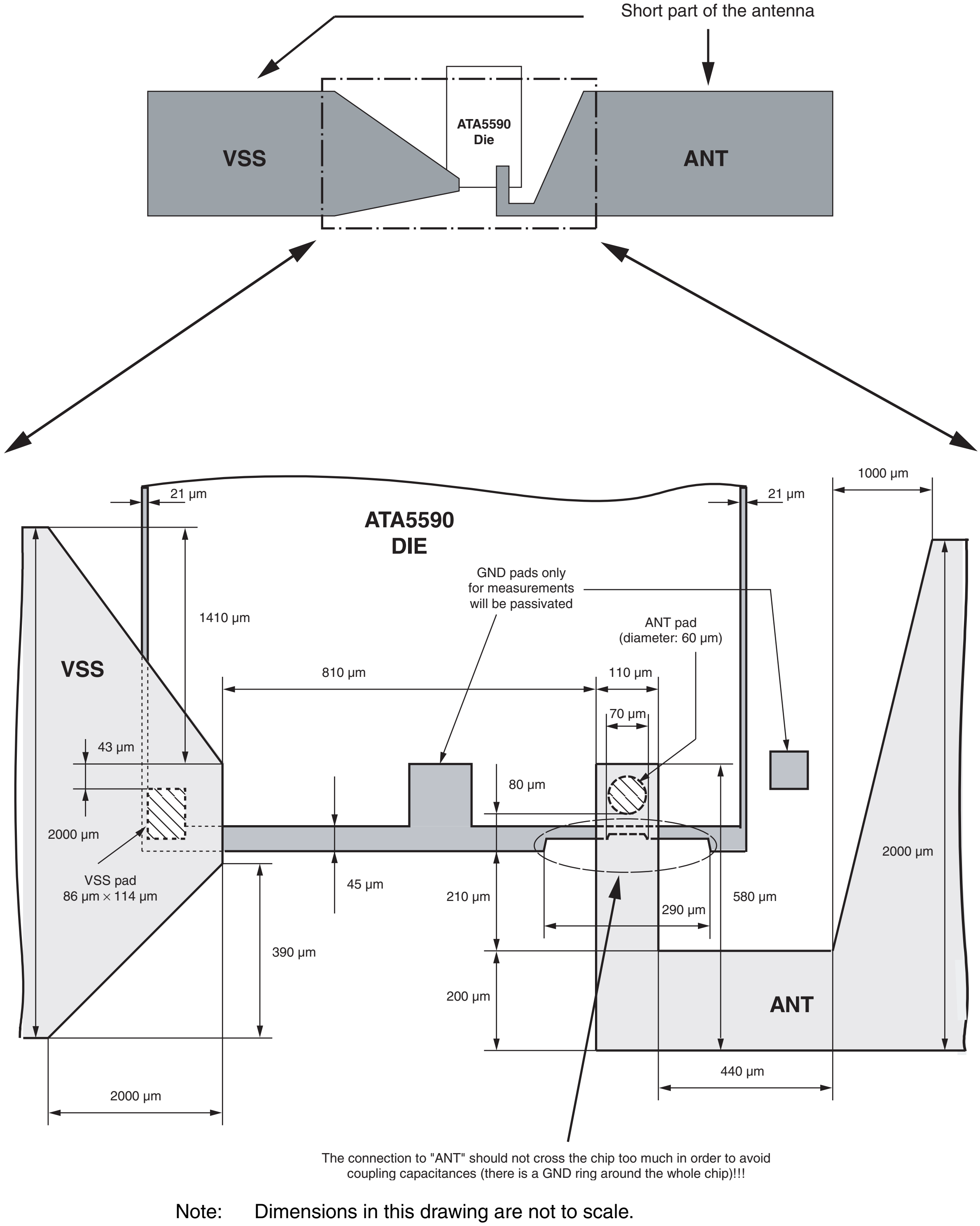
If the assembly tolerance is ±50 µm, the ANT pad should be connected as shown in Figure 2-3 on page 9. The antenna feeder to the ANT pad should have a width of 110 µm, thus ensuring a good connection in any case.

Figure 2-2. ANT Pad Connection Recommendation for an Assembly Tolerance ±50 µm and an Antenna Feeder Width of 110 µm



Note: The GND connection can cross the GND ring on a large surface without having any affect on the read distance.

Figure 2-3. Antenna Connection (Bottom View): Assembly Tolerance: ±50 µm

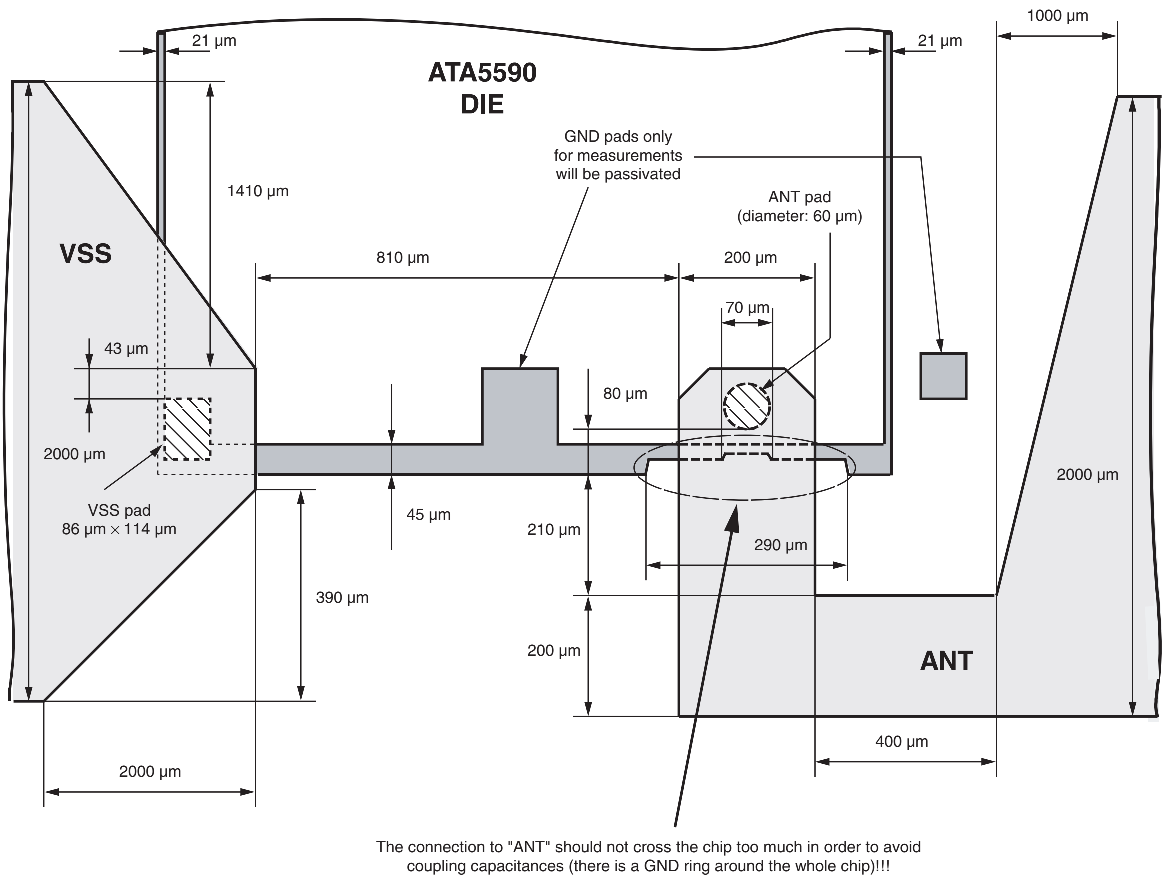


Note: Dimensions in this drawing are not to scale.

2.3 Antenna Connection Recommendation – Assembly Tolerance: ±50 µm

If the assembly tolerance is ±50 µm and the minimum antenna feeder is 200 µm, the ANT pad should be connected as shown in Figure 2-4, thus ensuring a good connection in any case.

Figure 2-4. Antenna Connection (Bottom View) – Assembly Tolerance: ±50 µm

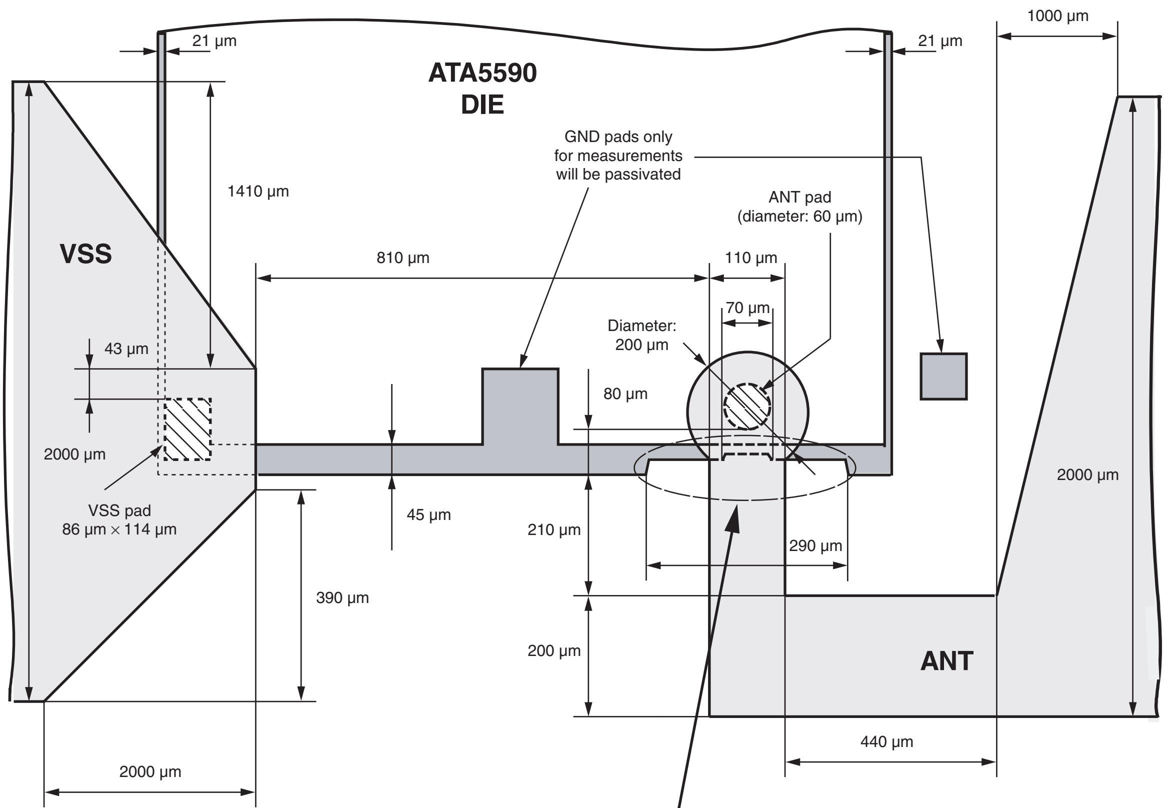


Note: Dimensions in this drawing are not to scale.

2.4 Antenna Connection Recommendation – Assembly Tolerance: ±100 µm

With this type of connection between ANT pad and antenna, the assembly tolerance can be increased from ±50 µm to ±100 µm.

Figure 2-5. Antenna Connection (Bottom View) – Assembly Tolerance: ±100 µm



Note: 1. Dimensions in this drawing are not to scale.

2. The antenna landing pad area of the ANT pad will be increased significantly, and, therefore, the parasitic capacitance will also be increased, resulting in a shift of the resonance frequency.

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